
Tecnologie del Linguaggio Naturale

Parte Prima

Lezione n. 04-1

> Sintassi: la competence

27 Febbraio 2026

Prologo

This is the cat that caught the rat that stole the cheese.

The man who wrote the book that you told me about

Aspects of the theory of syntax (1957)

N. Chomsky



Prologo

English Center-Embedding

A man that a woman loves

A man that a woman that a child knows loves

A man that a woman that a child that a bird saw knows loves

A man that a woman that a child that a bird that I heard saw knows
loves



Prologo

E venne il macellaio,
che uccise il toro,
che bevve l'acqua,
che spese il fuoco,
che bruciò il bastone,
che picchiò il cane,
che morse il gatto,
che si mangiò il topo
che al mercato mio padre comprò.



Alla fiera dell'est (1976)

A. Branduardi

Prologo

Elena e Maria odiano il latte e il caffè
rispettivamente

Outline

- Sintassi e grammatiche generative
- La gerarchia di Chomsky e il linguaggio naturale
- Mildly-Context Sensitive Languages, Tree Adjoining Grammars, Combinatory Categorical Grammars

Outline

- Sintassi e grammatiche generative
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NLP pipeline

Phonetics acoustic and perceptual elements

Phonology inventory of basic sounds (phonemes) and basic rules for combination, e.g. vowel harmony

Morphology how morphemes combine to form words, relationship of phonemes to meaning

Syntax sentence formation, word order and the formation of constituents from word groupings

Semantics how do word meanings recursively compose to form sentence meanings (from syntax to logical formulas)

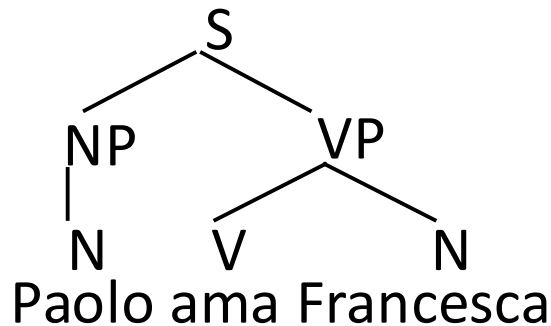
Pragmatics meaning that is not part of compositional meaning

Syntax and Semantics

Paolo ama Francesca



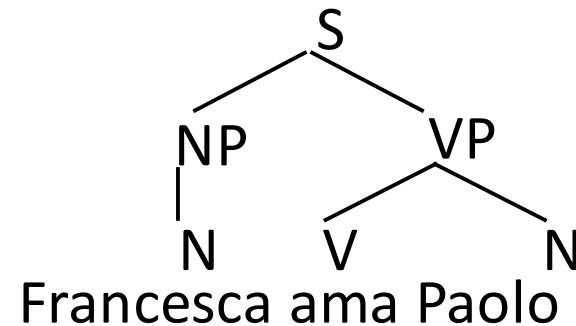
Syntactic Parsing



Francesca ama Paolo



Syntactic Parsing



Competence/Performance



"Linguistic theory is concerned primarily with an ideal speaker-listener, in a completely homogeneous speech-community, who know its (the speech community's) language perfectly and that it is unaffected by such **grammatically irrelevant conditions** as memory limitations, distractions, shifts of attention and interest, and errors (random or characteristic) in applying his knowledge of this language in actual performance."

Competence/Performance in CL

- **Competence = Grammatica Formale**
- **Performance = Algoritmo di Parsing**

Competence/Performance in CL

- Competence = Grammatica Formale
 - Performance = Algoritmo di Parsing
- > Prox. lezione: “Anatomia di un parser”

Rewriting Systems

Emil Post , Alan Turing

Rewriting rule



$$\psi \rightarrow \theta$$



Generative grammar



$$G=(\Sigma,V,S,P)$$

Σ = alphabet

$V = \{A,B,\dots\}$

$S \in V$

$P = \{\psi \rightarrow \theta, \dots\}$

Generative Grammars and Natural Languages

- Generative Grammars models the natural languages as a formal languages
- The derivation tree can model the **syntactic structure** of the sentence

Context-Free Grammars

$$G=(\Sigma,V,S,P) \quad A \rightarrow \beta$$

- Constituency
- Grammatical relations
- Subcategorization

Constituency

Context-Free Grammars Constituency

Hypothesis:

Constituent $\Leftrightarrow$ non terminal symbols V

Toy Grammar

$$G_4 = (\Sigma_4, \{S, NP, VP, V_1, V_2\}, S, P_4)$$

$$\Sigma_4 = \{I, Anna, John, Harry, saw, see, swimming\}$$

$$P_4 = \{S \rightarrow NPVP, VP \rightarrow V_1S, VP \rightarrow V_2,$$

$$NP \rightarrow I | John | Harry | Anna,$$

$$V_1 \rightarrow saw | see, V_2 \rightarrow swimming\}$$

Toy Grammar

S

$S \rightarrow NP VP$

$VP \rightarrow V_1 S$

$VP \rightarrow V_2$

$NP \rightarrow I | John | Harry | Anna$

$V_1 \rightarrow saw | see$

$V_2 \rightarrow swimming$

S

Toy Grammar

$S \Rightarrow NP VP$

$S \rightarrow NP VP$

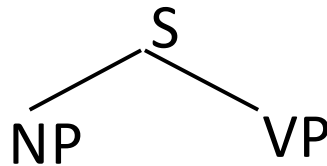
$VP \rightarrow V_1 S$

$VP \rightarrow V_2$

$NP \rightarrow I | John | Harry | Anna$

$V_1 \rightarrow saw | see$

$V_2 \rightarrow swimming$



Toy Grammar

$S \Rightarrow NP VP \Rightarrow I VP$

$S \rightarrow NP VP$

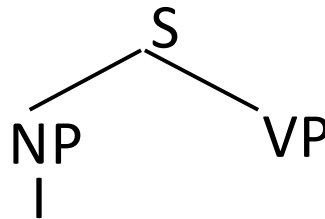
$VP \rightarrow V_1 S$

$VP \rightarrow V_2$

$NP \rightarrow I | John | Harry | Anna$

$V_1 \rightarrow saw | see$

$V_2 \rightarrow swimming$



Toy Grammar

$S \Rightarrow NP VP \Rightarrow I VP \Rightarrow I V_1 S$

$S \rightarrow NP VP$

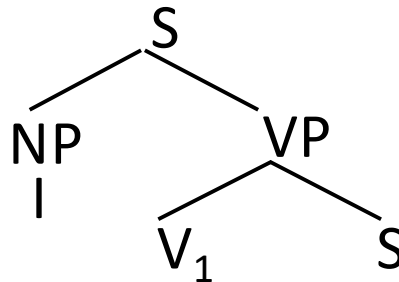
$VP \rightarrow V_1 S$

$VP \rightarrow V_2$

$NP \rightarrow I | John | Harry | Anna$

$V_1 \rightarrow saw | see$

$V_2 \rightarrow swimming$



Toy Grammar

$S \Rightarrow NP VP \Rightarrow I VP \Rightarrow I V_1 S \Rightarrow$

I saw S

$S \rightarrow NP VP$

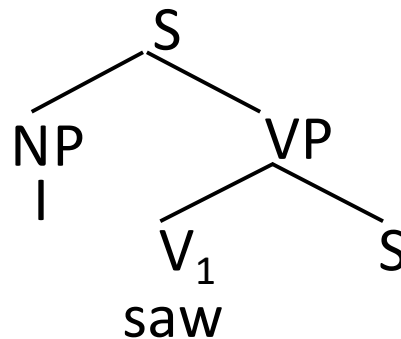
$VP \rightarrow V_1 S$

$VP \rightarrow V_2$

$NP \rightarrow I \mid \text{John} \mid \text{Harry} \mid \text{Anna}$

$V_1 \rightarrow \text{see} \mid \text{swimming}$

$V_2 \rightarrow \text{swimming}$



Toy Grammar

$S \Rightarrow NP VP \Rightarrow I VP \Rightarrow I V_1 S \Rightarrow$

$I \text{ saw } S \Rightarrow I \text{ saw } NP VP$

$S \rightarrow NP VP$

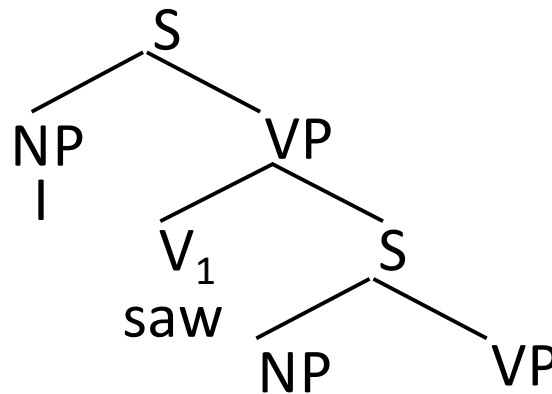
$VP \rightarrow V_1 S$

$VP \rightarrow V_2$

$NP \rightarrow I | \text{John} | \text{Harry} | \text{Anna}$

$V_1 \rightarrow \text{saw} | \text{see}$

$V_2 \rightarrow \text{swimming}$



Toy Grammar

$S \Rightarrow NP VP \Rightarrow I VP \Rightarrow I V_1 S \Rightarrow$

$I \text{ saw } S \Rightarrow I \text{ saw } NP VP \Rightarrow$

$I \text{ saw Harry } VP$

$S \rightarrow NP VP$

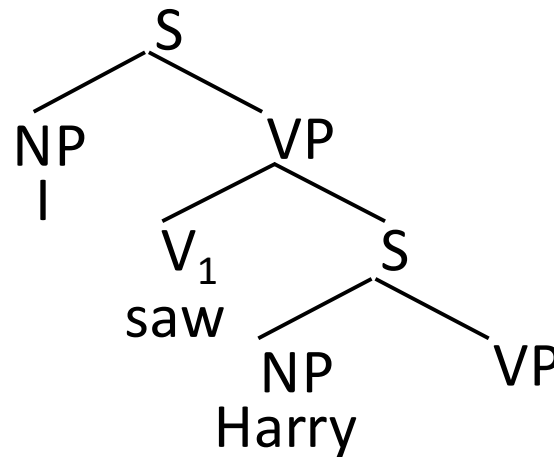
$VP \rightarrow V_1 S$

$VP \rightarrow V_2$

$NP \rightarrow I | \text{John} | \text{Harry} | \text{Anna}$

$V_1 \rightarrow \text{saw} | \text{see}$

$V_2 \rightarrow \text{swimming}$



Toy Grammar

$S \Rightarrow NP VP \Rightarrow I VP \Rightarrow I V_1 S \Rightarrow$

$I \text{ saw } S \Rightarrow I \text{ saw } NP VP \Rightarrow$

$I \text{ saw Harry } VP \Rightarrow$

$I \text{ saw Harry } V_2$

$S \rightarrow NP VP$

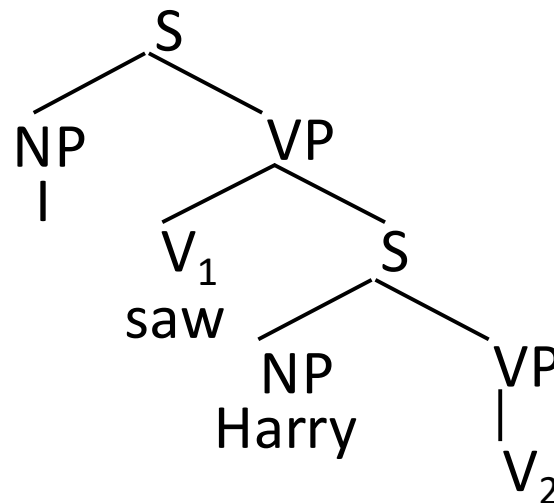
$VP \rightarrow V_1 S$

$VP \rightarrow V_2$

$NP \rightarrow I | \text{John} | \text{Harry} | \text{Anna}$

$V_1 \rightarrow \text{saw} | \text{see}$

$V_2 \rightarrow \text{swimming}$



Toy Grammar

$S \Rightarrow NP VP \Rightarrow I VP \Rightarrow I V_1 S \Rightarrow$

$I \text{ saw } S \Rightarrow I \text{ saw } NP VP \Rightarrow$

$I \text{ saw Harry } VP \Rightarrow$

$I \text{ saw Harry } V_2 \Rightarrow$

$I \text{ saw Harry swimming}$

$S \rightarrow NP VP$

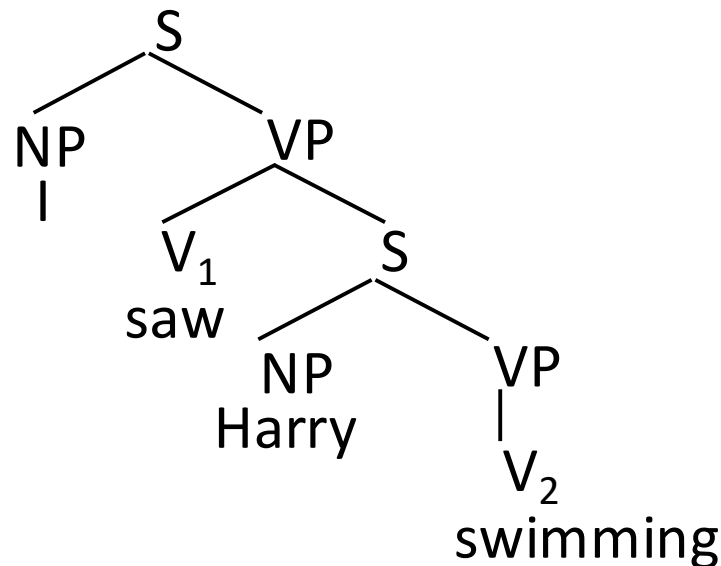
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$VP \rightarrow V_2$

$NP \rightarrow I | \text{John} | \text{Harry} | \text{Anna}$

$V_1 \rightarrow \text{saw} | \text{see}$

$V_2 \rightarrow \text{swimming}$



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Languages Chomsky hierarchy

Type 0

$\Psi \rightarrow \theta$

Context-sensitive

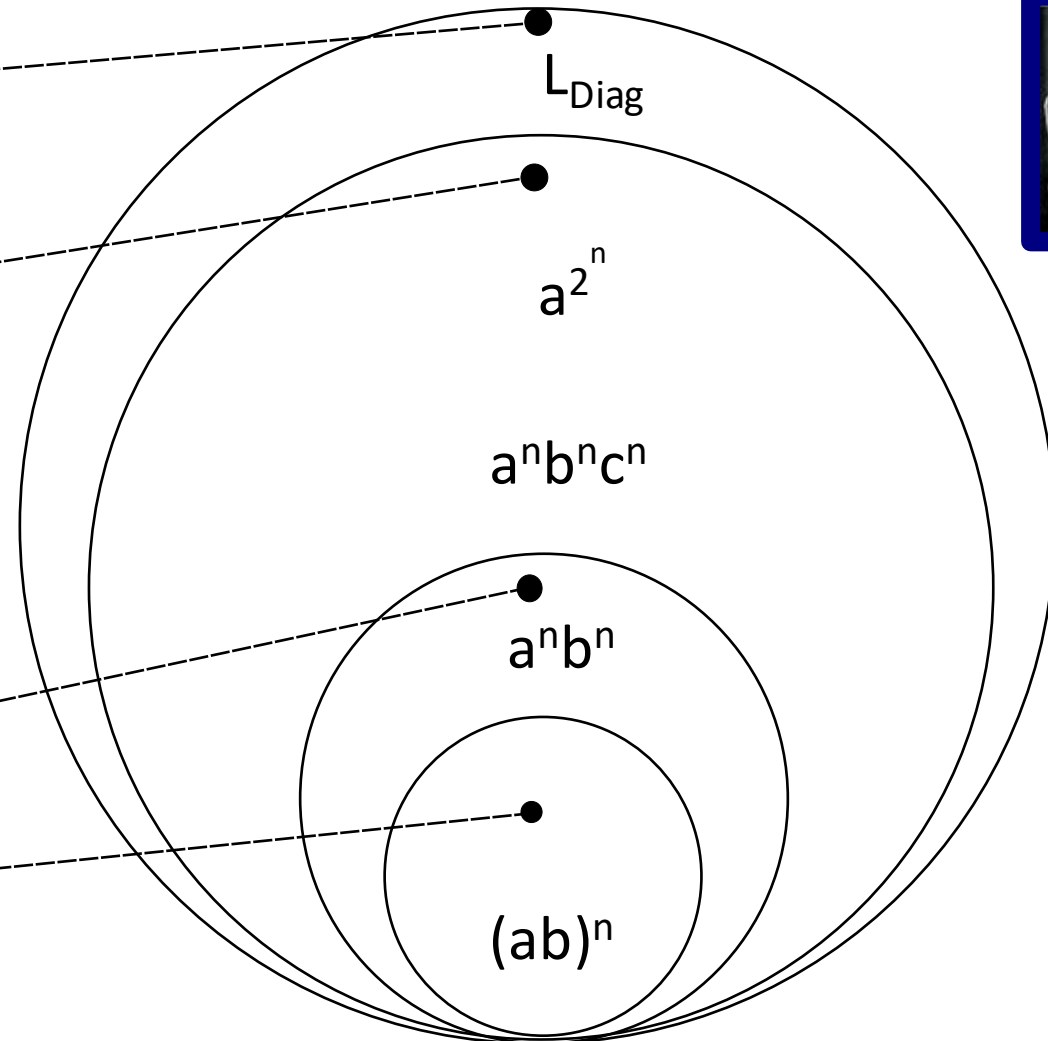
$Caa \rightarrow aaCa$

Context-free

$S \rightarrow aSb$

Linear

$A \rightarrow aB$



Swiss-German is NOT CF

(Shieber, 1985)



... das mer **em Hans** es huus **h'älfed** aastriiche

... that we Hans-Dat house-Acc helped paint

‘... that we helped Hans paint the house’

... das mer **d'chind** **em Hans** es huus **l'önd**
h'älfe aastriiche

... that we the childrenAcc HansDat houseAcc
let help paint

‘... that we let the children help Hans paint the
house’

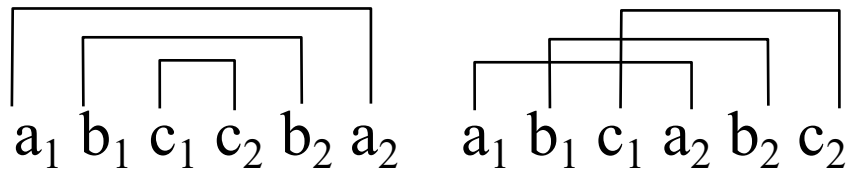
Outline

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Mildly context sensitive languages

[Joshi85]

- Include CFG languages
- Nested and cross-serial dependencies



- Polynomially parsable
- Constant growth property

Constant growth property

- **Definition** A language L is constant growth if there is a constants c_0 and a finite set of constant C such that for all $w \in L$ where $|w| > c_0$ there is a $w' \in L$ such that $|w| = |w'| + c$ for some $c \in C$.
- This property is the formal version of the linguistic intuition that the sentence belonging to a natural language can be built from a finite set of bounded structures using the same linear operations [Wei88].

Languages Chomsky hierarchy

Type 0

$\Psi \rightarrow \theta$

Context-sensitive

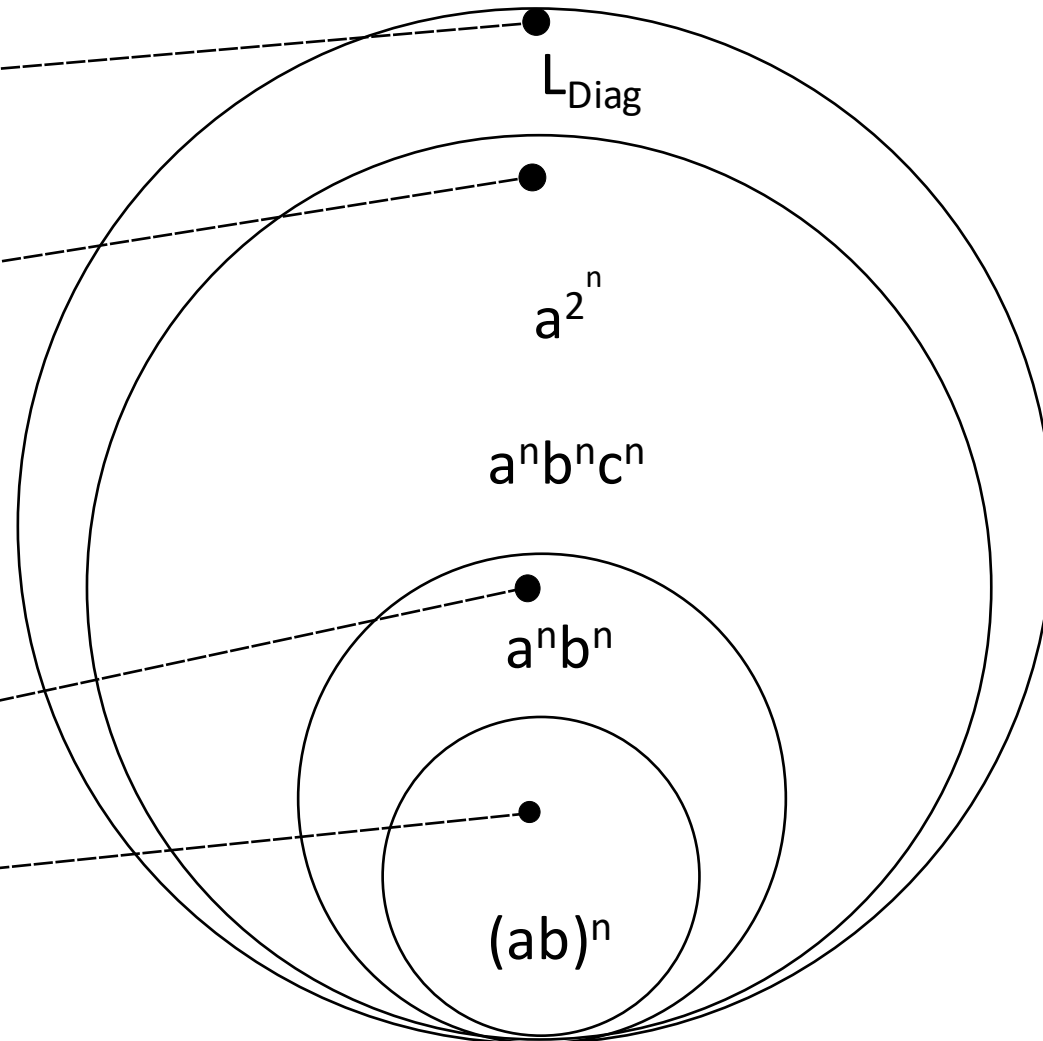
$Caa \rightarrow aaCa$

Context-free

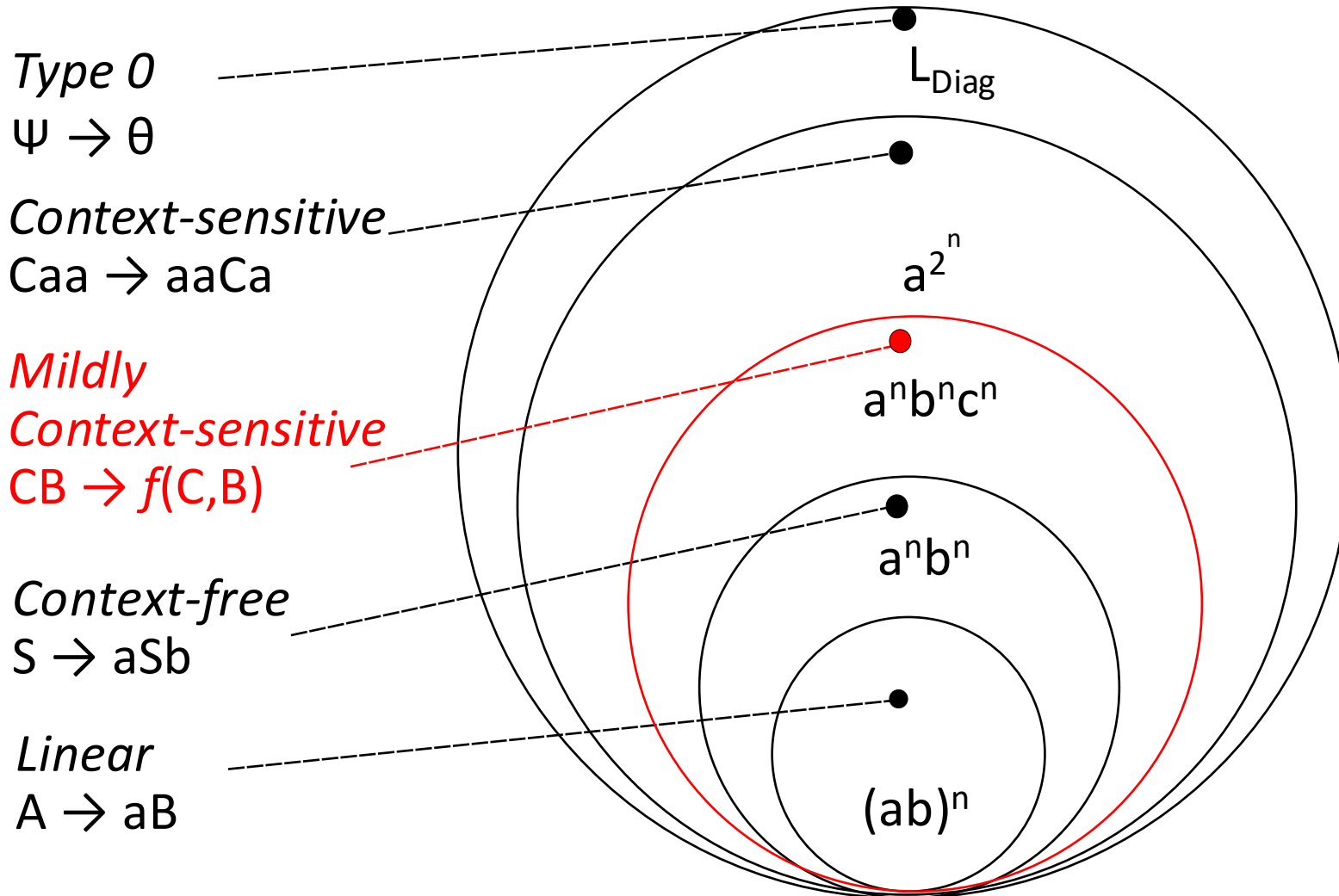
$S \rightarrow aSb$

Linear

$A \rightarrow aB$



Languages Chomsky hierarchy



MCL $\Leftrightarrow$ TAG, HG, LIG, CCG

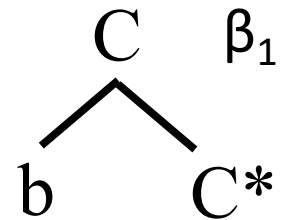
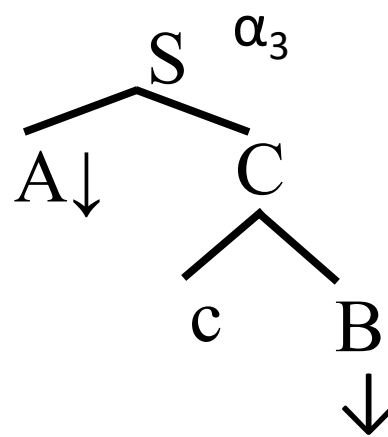
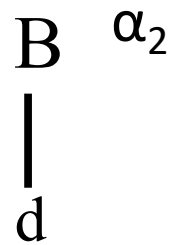
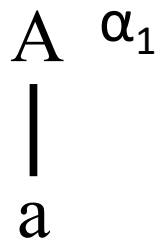
- Tree Adjoining Grammars (Joshi et al. 1975)
- Head Grammars (Pollard 1984)
- Linear Indexed Grammars (Gazdar 1985)
- Combinatory Categorical Grammars (Steedman 1985, Satta 2010)



-> elementary structures

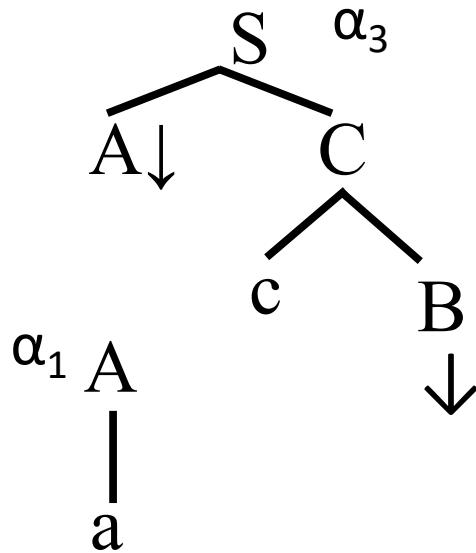
-> combination rules

Tree Adjoining Grammars



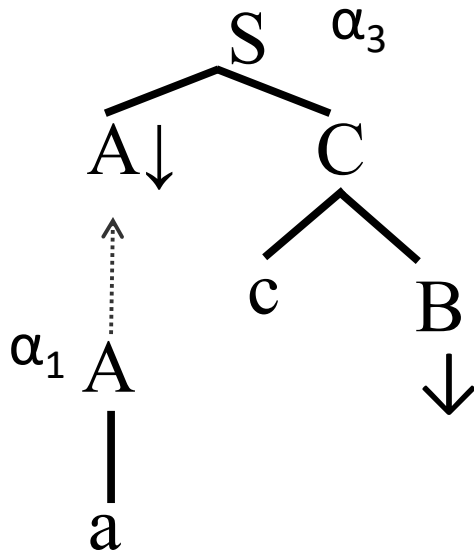
Elementary structures = **multilevel trees**

Tree Adjoining Grammars



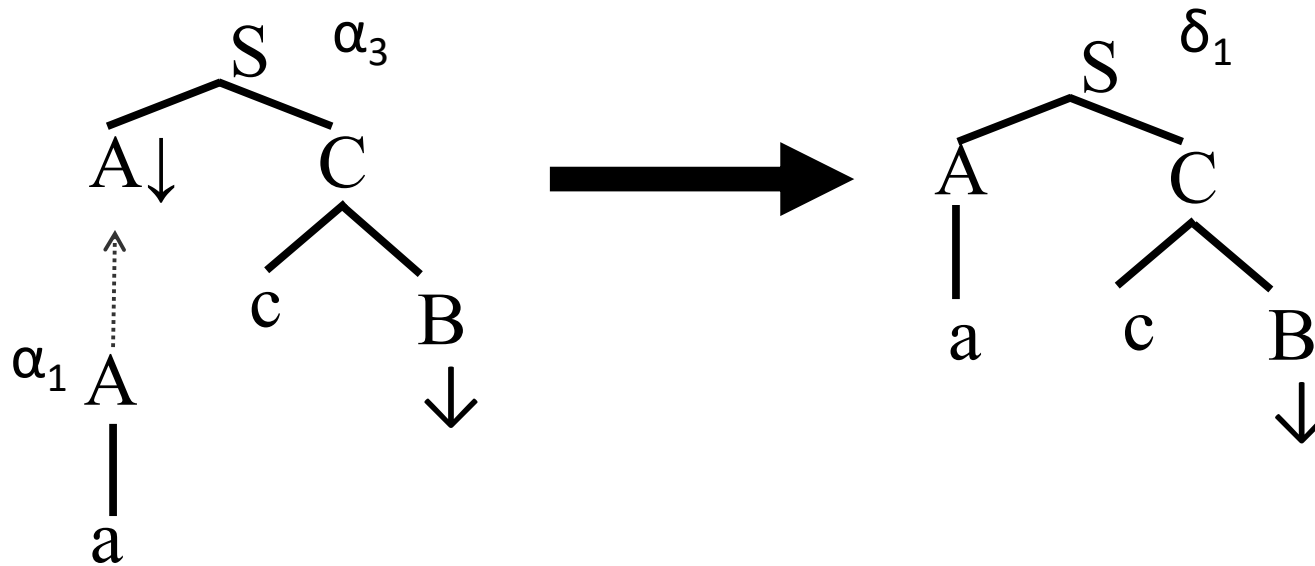
TAG operations: 1) substitution

Tree Adjoining Grammars



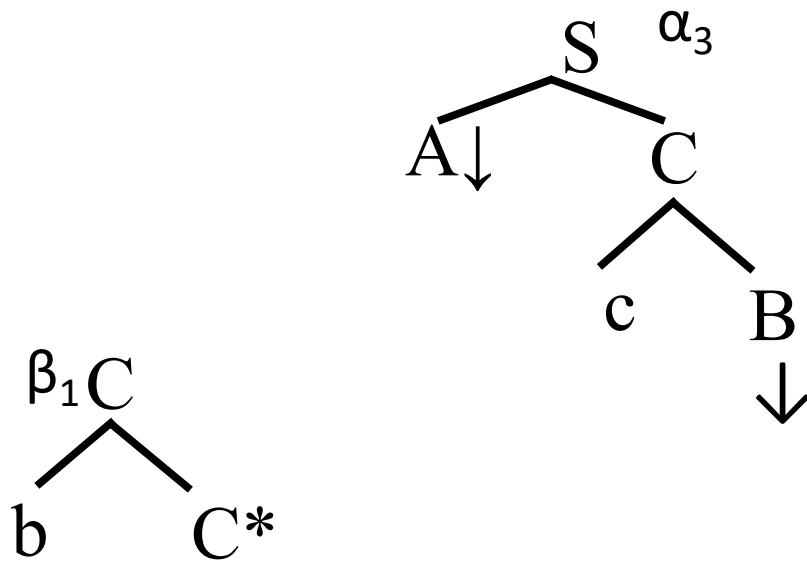
TAG operations: 1) substitution

Tree Adjoining Grammars



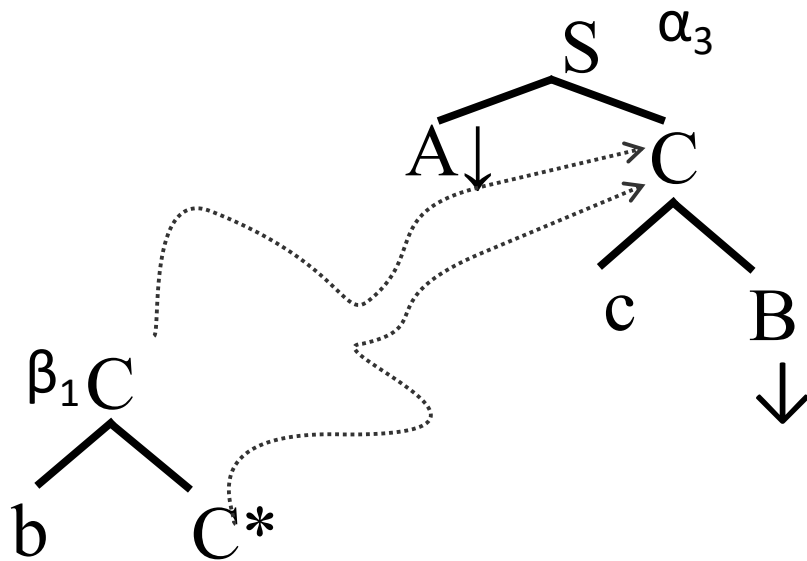
TAG operations: 1) substitution

Tree Adjoining Grammars



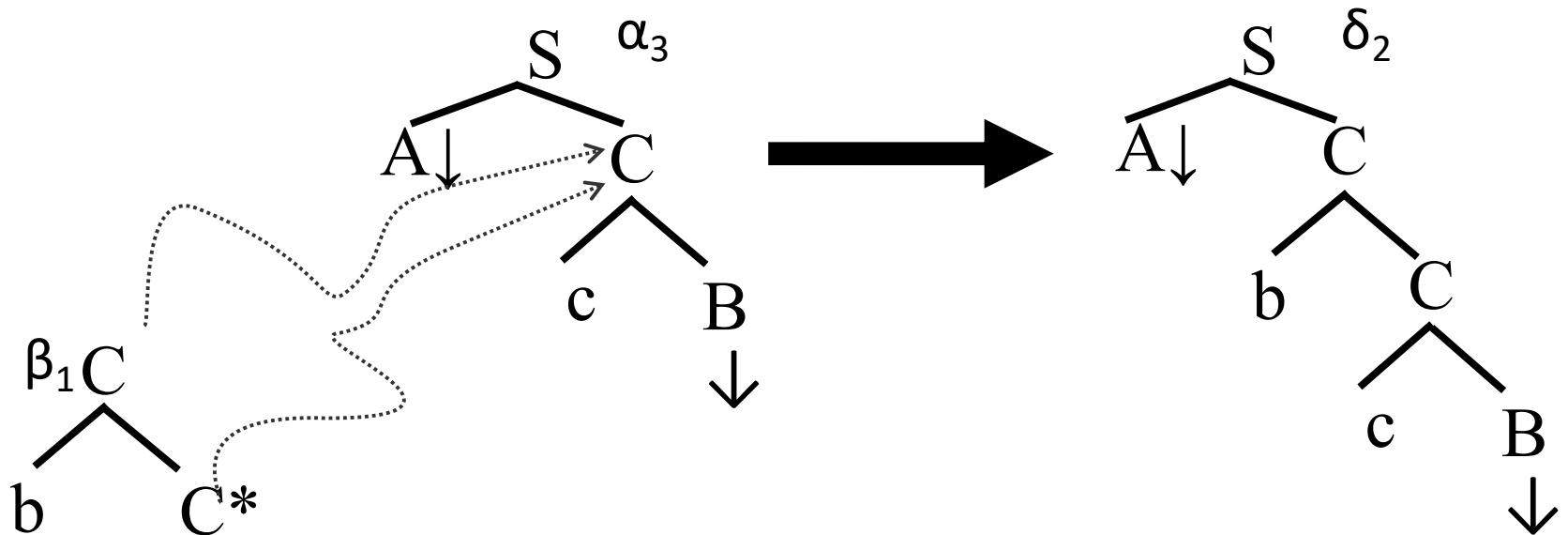
TAG operations: 2) adjoining

Tree Adjoining Grammars



TAG operations: 2) adjoining

Tree Adjoining Grammars



TAG operations: 2) adjoining

TAG and MCSL

- TAG properly contains all context-free languages (finitely ambiguous). Theorem (Schabes 1990)
- TAG is polynomially parsable}: $O(n^6)$
 - Embedded Push Down Automata, CKY (Vijay-Shanker 1987)
 - Left-to-right parser (Schabes 1990)

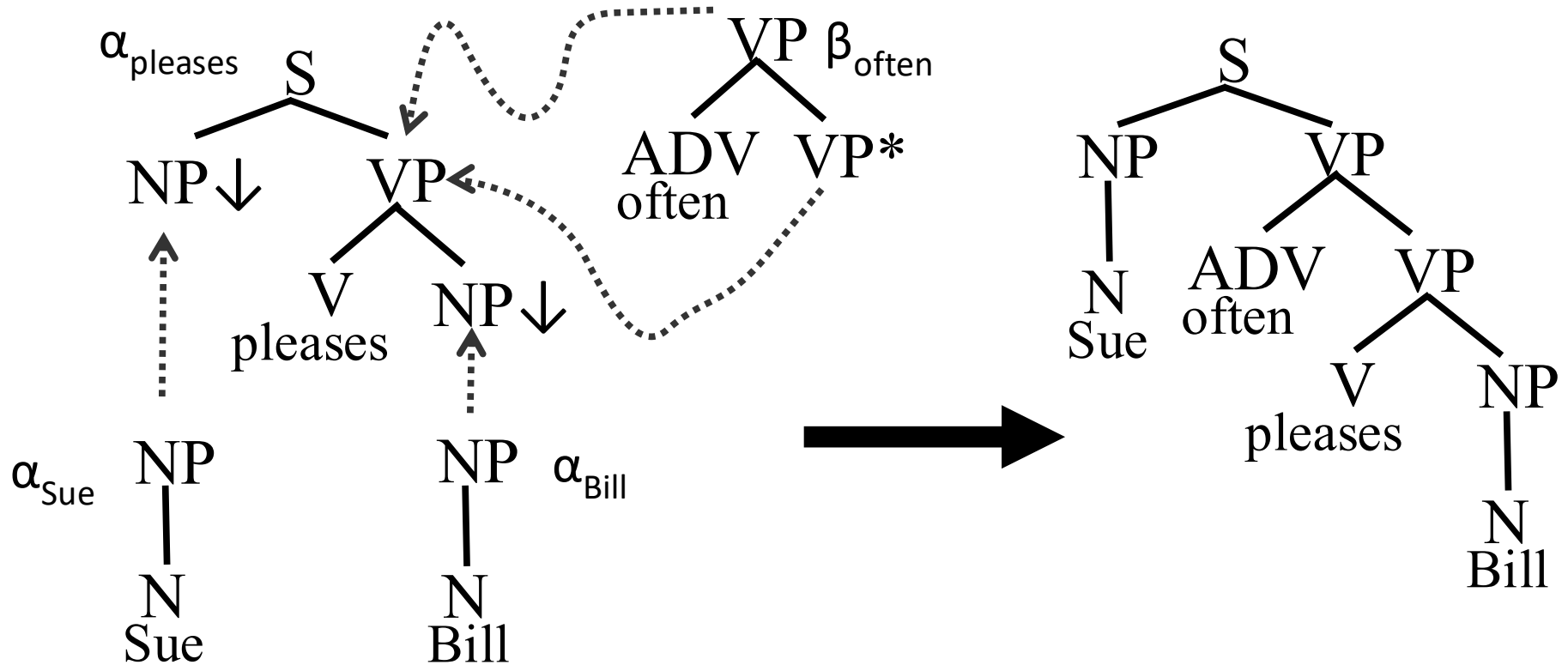
TAG and MCSL

- TAG captures only certain types of dependencies
 - **Cross-serial dependencies:** verb-raised analysis (Kroch Santorini 1991)
 - No mix-languages
- TAG has the constant--growth property: (Weir 1988)

Lexicalized Tree Adjoining Grammars

- Extended domain of locality
- Recursion Factorization by adjoining operation
- Lexicalization

LTAG



structures = multilevel trees

operations = substitution, adjoining

Combinatory Categorical Grammar

- Generative -> top-down
- Categoricali -> bottom-up

The Syntactic Process, *Mark Steedman*

Categories, Lexicon, Rules

Combinatory Categorical Grammar

- Generative -> top-down
- Categoricali -> bottom-up

Category

- > $A \subseteq C$, where A is a given set of atomic elements
- > $(X/Y), (X \backslash Y) \in C$, if $X, Y \in C$

CCG

- **Lexicon**

- Paolo : NP
- Francesca : NP
- amare : $(S \backslash NP) / NP$

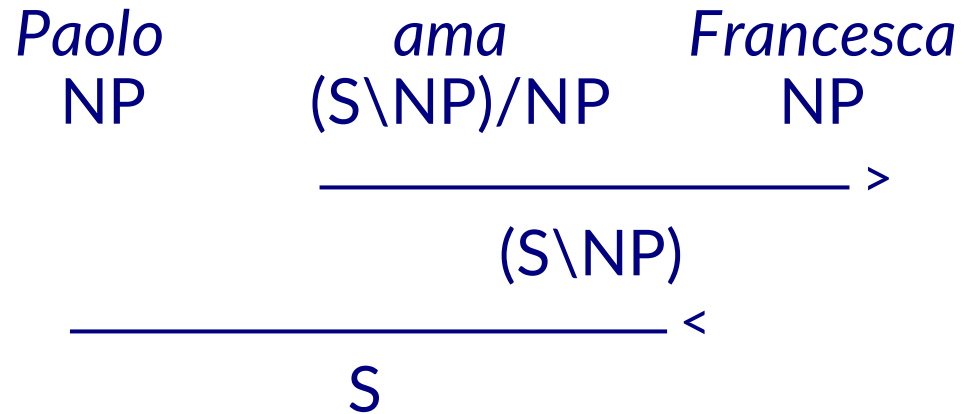
- **Rules**

- $X/Y \ Y \Rightarrow X \quad \rightarrow$
- $Y \ X \backslash Y \Rightarrow X \quad \leftarrow$
- ...

CCG

- Lexicon

- Paolo : NP
- Francesca : NP
- amare : (S\NP)/NP



- Rules

- $X/Y \ Y \Rightarrow X$ $->$
- $Y \ X \backslash Y \Rightarrow X$ $-<$
- ...

CCG

- Coordinazione: $X \text{ CONJ } X \Rightarrow X$

